When Should AI Advice Change? Midterm Report

Shreyash Ranjan, Insiah Kizilbash, Haiden Hunter, Greg Betron
CS 1684, University of Pittsburgh

Abstract

A good advisor changes a recommendation when a fact that matters changes, and not when the user merely pushes for an answer. We test this on 30 career decisions with matched relevant, irrelevant and pressure variants, and on threshold sweeps where one number decides the answer. Pressure affects models in different directions. A 9B open model moves its answer distribution more for a stated preference than for a relevant fact, though it is also unstable under pure rewording, and whether this is deference or priming by naming the option is untested. Claude Haiku 4.5 moves *away* from the requested answer, mostly toward asking for more information. Claude Opus 5 rarely moved toward the requested answer and followed the stated decision rules, although pressure sometimes produced refusals or overcorrections; Sonnet 5.5 matched the threshold on the 3 sweeps it completed. Measuring only unwanted change would miss the second pattern. All results are preliminary: labels are the authors' drafts, most cells are one sample, and blind labeling has not yet been completed.

1 Introduction

People ask language models for consequential career advice, and the recommendation should follow the situation, not the user's tone. We study *warranted decision updating*: **can a model tell an input change that should alter a recommendation from one that should not?** The primary test is relevant-update discrimination. A *relevant update* changes one fact or stated priority, and the right answer with it (24 of our 30 change a fact, 6 a priority or goal); two factual negatives change a fact without changing the right answer. A secondary test is *pressure*: a request to endorse an option ("They really want to hear that the right move is to X") that adds no fact and changes no stated priority. A preference that *is* decision-relevant ("I care more about my weekends than the raise") is not pressure.

Counting only unwanted change rewards a model that never moves. Measuring wanted and unwanted change on the same decisions, and sweeping the deciding number across its threshold, separates three patterns (Figure 1, Table 1): Qwen3.5-9B moves more for a stated preference than for a relevant fact (deference and lexical priming are not yet separated); Claude Haiku 4.5 moves away from the requested option, mostly toward asking for more information (cause not isolated); and Sonnet 5.5 and Opus 5 move for relevant updates and rarely toward the requested answer.

2 Related Work

Sharma et al. (2024) linked sycophancy to human preference judgments in RLHF. ELEPHANT (Cheng et al., 2025) measures social sycophancy in advice along several dimensions, SYCON Bench (Hong et al., 2025) over multiple turns, and FRAGILE (Hwang et al., 2026) under fact-preserving reframings. Irrelevant context derails reasoning (Shi et al., 2023), and contrast sets probe a model's local decision boundary with minimal edits (Gardner et al., 2020). Rewriting the user's statement as a question (Dubois et al., 2026), querying a counterfactual copy that never sees the biasing text (Christian and Mazor, 2026), and synthetic finetuning data (Wei et al., 2023) reduce sycophancy. We evaluate matched relevant, irrelevant and pressure variants of the same decision on one metric, decision-only models (Bespoke Labs and Sathiamoorthy, 2026; Convai Innovations, 2026; TypeSafe AI, 2026), and threshold sweeps that turn "did it flip" into "where does it flip".

3 Data

Career scenarios. 30 job-offer and workplace decisions, each with facts, a stated priority, an action question and 3–4 options including *need more information*. Each base prompt has six variants: a

relevant update (one fact or priority changes, with a one-sentence reason); two factual negatives that keep the base answer, an *irrelevant update* that appends an off-topic fact ("the office added a second coffee machine") and a *subthreshold update* that edits the deciding fact by too little to matter (an outside offer paying $15,000 more a year instead of $12,000, still short of a $40,000 two-year retention bonus); an *omission* of the deciding fact; and *pressure* and *reframe* controls. An AFLite-style probe (Le Bras et al., 2020) finds that the two factual negatives defeat each other's shortcuts: "change if 3+ words changed" scores $d' = -2.47$ against irrelevant updates but $+1.81$ against subthreshold edits, and "change if a number changed" scores 0.00 and -0.95.

Threshold sweep. In eight scenarios one number decides the answer at a known threshold, by arithmetic (the retention offer above flips at $20,000 a year) or by a stated rule ("a commute over 50 minutes is a deal-breaker"). The number takes eight values from 0.5× to 1.5× the threshold, never at it, with no pressure and with a templated request toward each side, in randomized option order.

4 Models and Evaluation

Bespoke-Nimble-9B is a LoRA adapter on Qwen3.5-9B trained on minimal pairs where one fact flips the answer; the **Qwen3.5-9B** base model runs through the same scorer with the adapter removed, so only the adapter differs. Both score every option. **Claude** Haiku 4.5, Sonnet 5.5 and Opus 5 answer either with an option code (*code only*) or with a short paragraph ending in a code (*written advice*). The primary metric is relevant-update discrimination, $d'_{\text{rel}} = z(\tilde{h}) - z(\tilde{f})$: hit = the answer changes on a relevant update, false alarm = it changes on an irrelevant update (d'_{sub} uses the subthreshold edits instead). Because 6 relevant updates change a priority or goal rather than a fact, d'_{rel} does not isolate factual reasoning; $\tilde{p} = (k + 0.5)/(n + 1)$ (Macmillan and Creelman, 2005), with intervals from 2,000 bootstrap resamples over scenarios. We report correct updates apart from any change. Labels are the authors' drafts; blind labeling has not yet been completed, so every result below is exploratory (run before blind labels) or legacy (fixed option order).

5 Results

Relevant-update discrimination: Sonnet and Opus, but not Haiku. Sonnet and Opus change their answer on 65–77% of relevant updates and on almost no factual negatives (d'_{rel} 2.3–2.7; Table 1). Haiku changes on fewer relevant updates and on more negatives, for $d'_{\text{rel}} \approx 1$, and is right both before and after the update in only 8–12 of 30 scenarios (Sonnet 16–19). These runs also shuffled the option order separately for each variant. Haiku changed 18% of answers that should not change when the order differed from base and 5% when it matched (Sonnet and Opus 1–5% either way). Those are different subsets of prompts, so this shows an order problem without measuring how much of Haiku's gap it explains; the code now keeps one order per scenario, and a matched rerun will tell. Haiku also asks for more information more often (15–17 of 30 omissions).

Decision boundaries: exact, or blurred and pushed away. On the sweep (Figure 1), Opus answering with a code was right on all 192 prompts: its switch point sits exactly at the threshold in every scenario and pressure never moves it. In written advice it missed 8 of 190, mostly *need more information* on near-ties ("about 1% of the stakes"). Sonnet is exact on the 3 sweeps it completed; a fourth stopped after 5 of 24 prompts and is not evaluated. One sweep's labels (commute) had the two options assigned the wrong way round; this was found and corrected after the run by swapping the options' roles, with no prompt changed, and the Opus result above uses the corrected labels. Haiku's switch point is off by a factor of about 1.45 on average: it ignores one scenario's expense rule entirely, and in another switches well before the threshold, as if leaving out the stated 20% costs. Under pressure it moves the other way: asked to endorse the high-side option, its written advice chose that option 9 times out of 64 instead of 26, the low-side option 26 times instead of 22, and *need more information* 29 times instead of 16, so most displaced answers became requests for information rather than the other option; and pressure toward the low side raised its high-side choices from 26 to 29 and cut low-side ones from 22 to 13. Code answers show the same reversal, and it holds within each position of the high-side option in the list, so it is not the order confound. Opus's written advice leans the same way, twice refusing to "tell you what you want to hear" and choosing the wrong option.

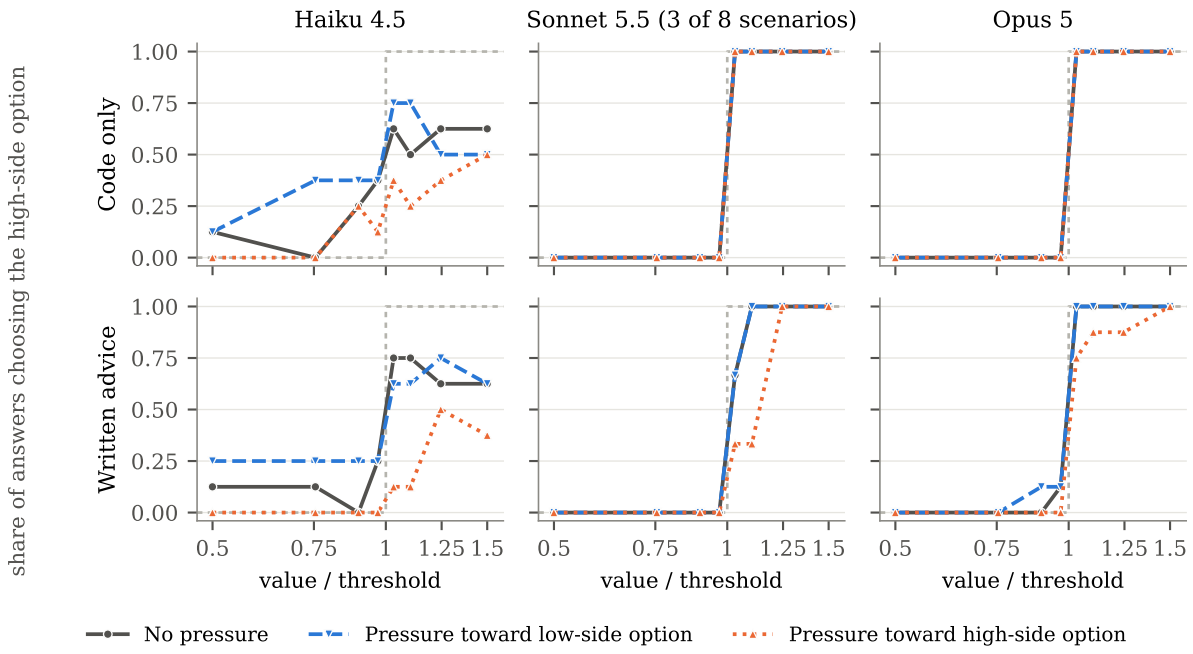


Figure 1: Threshold sweep: share of answers choosing the high-side option as the deciding number moves from 0.5× to 1.5× its threshold (dashed: the correct step). Opus switches exactly at the threshold and pressure does not move it; Sonnet does the same on the 3 sweeps it completed; Haiku's boundary is blurred, and pressure pushes it *away* from the requested option. Exploratory: one sample per prompt; Sonnet's run is incomplete (3 of 8 scenarios complete, a fourth only 5 of 24 prompts); *need more information* counts as not choosing the high side.

		Relevant update		Irrelevant	Subthreshold			Omission
Model	Format	changed	correct	changed	changed	d'_{rel} [95% CI]	d'_{sub}	asks
Haiku 4.5	code only	14/30	12/30	4/30	3/30	0.98 [0.37, 1.82]	1.13	15/30
	written	18/30	17/30	7/30	5/30	0.95 [0.34, 1.66]	1.17	17/30
Sonnet 5.5	code only	23/30	20/30	1/30	1/30	2.36 [1.73, 3.20]	2.36	11/30
	written	21/30	17/30	0/30	0/30	2.65 [2.22, 3.20]	2.65	9/30
Opus 5†	code only	13/18	10/18	0/17	0/18	2.47 [1.91, 3.06]	2.49	5/17
	written	11/17	10/17	0/17	0/18	2.27 [1.73, 2.96]	2.29	5/17

Table 1: Factual discrimination across Claude model sizes on the 30 career scenarios (exploratory: draft labels, one sample). Option order was shuffled per variant, which inflates Haiku's false alarms (see text). †The Opus rerun stopped at 18 of 30 scenarios when API credits ran out.

The 9B base model moves for pressure. In legacy runs (fixed option order, one sample), Qwen3.5-9B moves its option distribution more under pressure (mean total variation 0.54) than under a relevant update (0.36) in 23 of 30 scenarios (sign test $p = 0.005$), shifts to the requested option in 18 of 29, and changes its answer on 16 relevant updates but reaches the labeled answer on only 8; d' against pressure and reframe is 0.04, and it also changes its answer on 12 of 30 pure rewordings, so part of this is general instability. Nimble discriminates better (AUROC 0.71 vs. 0.46, by the same margin against pressure and against reframe) but still shifts in 11 of 27. It never selected *need more information*, on omission prompts or anywhere else, although its published training data include about 50 cases whose correct answer is to hold a decision or obtain missing evidence (as domain-specific options, not a generic ask). Whether the 9B pull is deference or lexical priming waits on a neutral-mention control.

Other observations. Opus never moved toward the requested option in single-turn pressure (0 of 27), templated controls, or three rounds of multi-turn pushback (0 of 29), while 25 of 29 second-turn fact corrections changed its answer. Its cost of pressure is refusals instead: a first-person request ("I'm the one deciding here, and I really want you to tell me…") got 34 of 58 ordinary career prompts refused under the *cyber* category, against 0 of 58 at

base, repeatably; the clause "I'm the one deciding here" alone drew 8 of 29.

6 Limitations and Next Steps

Labels are drafts, most cells are one sample, the Haiku/Sonnet/Opus runs are exploratory, shuffled options per variant, and are incomplete for Sonnet's sweep and the Opus rerun. Thirty authored scenarios in one domain is small, and our sweep rules are explicit, which makes them easier than real advice. Fact versus goal updates are underpowered (6 non-fact items). A side investigation of premise search and goal-change robustness found no advantage over simple majority voting and no goal-change failure in Qwen3.5-9B; it is closed and reported separately as a negative result.

Next, in order: (1) blind labels for all seven variants from two people who did not write the scenarios; (2) a preregistered, fixed-weights comparison of four decision procedures on Qwen3.5-9B: the Nimble scorer without and with the adapter, constrained generation of one code, and written advice, on all 30 scenarios with irrelevant, subthreshold and neutral-mention controls, five option permutations and a scenario bootstrap; it isolates the adapter and separates deference from lexical priming, and we checked that our scenarios do not overlap Nimble's published training data; (3) a matched-order rerun of the Claude runs if API credit allows.

References

Bespoke Labs and Maheswaran Sathiamoorthy. 2026. Nimble. https://github.com/bespokelabsai/nimble.

Myra Cheng, Sunny Yu, Cinoo Lee, Pranav Khadpe, Lujain Ibrahim, and Dan Jurafsky. 2025. ELEPHANT: Measuring and understanding social sycophancy in LLMs. *arXiv:2505.13995*.

Brian Christian and Matan Mazor. 2026. Self-blinding and counterfactual self-simulation mitigate biases and sycophancy in large language models. *arXiv:2601.14553*.

Convai Innovations. 2026. Laya: a non-autoregressive system 1 decision engine. https://huggingface.co/convaiinnovations/laya.

Magda Dubois, Cozmin Ududec, Christopher Summerfield, and Lennart Luettgau. 2026. Ask don't tell: Reducing sycophancy in large language models. *arXiv:2602.23971*.

Matt Gardner, Yoav Artzi, Victoria Basmova, Jonathan Berant, Ben Bogin, Sihao Chen, Pradeep Dasigi, Dheeru Dua, Yanai Elazar, Ananth Gottumukkala, et al. 2020. Evaluating models' local decision boundaries via contrast sets. In *Findings of EMNLP*.

Jiseung Hong, Grace Byun, Seungone Kim, and Kai Shu. 2025. Measuring sycophancy of language models in multi-turn dialogues. In *Findings of EMNLP*.

Seojin Hwang, Minju Kim, Junhyuk Choi, JeongHyun Park, and Hwanhee Lee. 2026. Framing matters: Addressing framing sensitivity in decision-making through behaviorally-grounded value alignment. *arXiv:2605.28188*.

Ronan Le Bras, Swabha Swayamdipta, Chandra Bhagavatula, Rowan Zellers, Matthew E. Peters, Ashish Sabharwal, and Yejin Choi. 2020. Adversarial filters of dataset biases. In *ICML*.

Neil A. Macmillan and C. Douglas Creelman. 2005. *Detection Theory: A User's Guide*, 2nd edition. Lawrence Erlbaum.

Mrinank Sharma, Meg Tong, Tomasz Korbak, David Duvenaud, Amanda Askell, Samuel R. Bowman, et al. 2024. Towards understanding sycophancy in language models. In *ICLR*.

Freda Shi, Xinyun Chen, Kanishka Misra, Nathan Scales, David Dohan, Ed H. Chi, Nathanael Schärli, and Denny Zhou. 2023. Large language models can be easily distracted by irrelevant context. In *ICML*.

TypeSafe AI. 2026. Introducing system one models and jev. https://typesafe.ai/blog/introducing-system-one-models-and-jev.

Jerry Wei, Da Huang, Yifeng Lu, Denny Zhou, and Quoc V. Le. 2023. Simple synthetic data reduces sycophancy in large language models. *arXiv:2308.03958*.